Weakly nonlinear scalar scattering from a Kerr black hole

KerrScattering Collaboration[1]

[1] *KerrScattering Project*

We investigate the weakly nonlinear scattering problem for a self-interacting complex scalar field in Kerr spacetime. The calculation follows the same flux-based logic as the Schwarzschild nonlinear Green-function construction: the nonlinear correction to the transmission and reflection coefficients is read from the first-order change of the energy flux at the future event horizon and at future null infinity. The Kerr extension replaces spherical partial waves by scalar spheroidal harmonics, introduces the co-rotating horizon frequency $p = \omega - m\Omega_H$, and turns the cubic source into a target-channel matrix. For the main comparison with the Schwarzschild paper we specialize to axisymmetric modes $m = 0$ with $a = 0.5$ and compute the self-channel coefficients for $\ell = 0, 1, 2$. We find Breit-Wigner-type peaks in the nonlinear transmission coefficient, with peak locations tracking the frequency scale set by the real parts of the corresponding Kerr scalar quasinormal modes. Over finite high-frequency windows, the coefficients are well fit by an exponential tail with effective temperatures within 3.2–13.6% of the Kerr Hawking temperature; this is a finite-window numerical diagnostic, not a claim of an exact thermal law. The reported nonlinear coefficients are derivatives at $\epsilon = 0$ in the fixed-incident normalization, rather than complete finite-coupling fluxes. representative non-axisymmetric calculation also resolves the Kerr-specific target-channel matrix generated by the spheroidal source projection. A bounded non-axisymmetric scan at $\ell = m = 2$ crosses the superradiant threshold $m\Omega_H$ and resolves the corresponding sign change in the signed horizon flux coefficients on the accepted points. The low-frequency Kerr data also expose a practical difference from the Schwarzschild reduced-field calculation: the reflected first-order flux is more cancellation limited, so the flux balance is used as the numerical precision diagnostic. On the accepted frequency points, the largest first-order balance residuals are 4.3×10^{-12}, 2.2×10^{-12}, and 1.2×10^{-12} for $\ell = 0, 1, 2$, respectively, while the measured low-frequency transmission slopes are 2.50, 5.48, and 8.74.

I. INTRODUCTION

We use geometric units $G = c = 1$, metric signature $(-, +, +, +)$, and set $M = 1$ in the numerical results unless stated otherwise. The Fourier convention is fixed by the mode dependence $e^{-\mathrm{i}\omega t + \mathrm{i}m\phi}$ used below.

Scattering is a foundational probe of black-hole spacetimes. The linear problem has been studied extensively for scalar, electromagnetic, and gravitational waves in Schwarzschild and Kerr backgrounds, using partial-wave, complex-angular-momentum, and related frequency-domain methods [1–6]. In a standard scattering experiment one sends in a monochromatic mode from past null infinity and compares the incident flux with the reflected flux at future null infinity and the absorbed flux at the future event horizon. The resulting transmission and reflection coefficients are natural complements to quasinormal-mode spectroscopy [7]. This point has become sharper because quasinormal spectra can be sensitive to small deformations of the effective potential [8–10], while scattering data, greybody factors, and related response functions can remain stable observables [11–15].

The nonlinear scattering problem is less developed. Time-domain studies have examined nonlinear perturbative and fully nonlinear scattering from past to future null infinity [16, 17], and recent work on nonlinear dynamics in general relativity has studied frequency-dependent scattering and mode generation [27]. The Schwarzschild calculation that we use as a template instead keeps the flux definition of scattering coefficients explicit: it considers a weak scalar self-interaction, solves the linear homogeneous problem, uses the nonlinear source to construct a first-order Green-function correction, and defines nonlinear scattering coefficients from the interference terms in the energy flux. This is the key point: the first-order field is not a new independent linear scattering experiment; it changes the physical boundary fluxes by interfering with the zeroth-order wave.

The scalar model is not a gravitational self-force calculation, but its bookkeeping is closely related to the sourced perturbative hierarchies used in extreme-mass-ratio inspiral (EMRI) theory: a lower-order field generates a higher-order source, mode coupling must be projected consistently, and observable fluxes provide an important global check [18–20]. We use this analogy only to motivate the workflow; no claim about a gravitational waveform or a self-force observable is made here. Frequency-domain calculations for a scalar charge on Kerr, generic bound Kerr geodesics, and the associated regularization parameters make the additional worldline and singular-field structures explicit [21, 23, 24]. Effective-source and extended-particular-solution methods provide a related scalar prototype for controlling frequency-domain reconstruction, but they retain a singular worldline source and regularization step absent here [22]. Recent second-order self-force and metric-reconstruction work further shows why those structures cannot be replaced by the finite cubic source used here [25, 26].

This work carries the same chain to the scalar Teukolsky equation in Kerr spacetime with spin weight $s = 0$.

The aim is to keep the observable definition and the flux logic of the Schwarzschild work unchanged while replacing the radial and angular equations by their Kerr counterparts. Rotation produces three changes. First, the horizon flux is proportional to $p = \omega - m\Omega_H$, which is the origin of superradiant sign changes [28, 29]. Second, spherical harmonics are replaced by spheroidal harmonics. Third, the covariant scalar source contains $\Sigma = r^2 + a^2\cos^2\theta$, so a single incident mode sources several target spheroidal channels.

The paper is organized in the same order as the Schwarzschild calculation, so that each Kerr step can be compared directly with its spherical counterpart. Section II defines the nonlinear scattering problem and the flux coefficients. Section III describes the pseudo-spectral homogeneous solver, the Green-function construction, the balance law, the accuracy diagnostics, and the low- and high-frequency diagnostics. Section IV gives the main frequency-domain numerical result: Kerr nonlinear transmission curves for $\ell = 0, 1, 2$, fitted resonance parameters, and high-frequency temperatures. The main physical frequency sweep is deliberately restricted to an axisymmetric Kerr sample, $a = 0.5$ and $m = 0$. This restriction makes the comparison with the Schwarzschild flux construction as direct as possible. Non-axisymmetric self-channel runs, including a bounded superradiant check, appear separately in Sec. IV; they are not folded into the axisymmetric production curves.

II. NONLINEAR SCATTERING PROBLEM IN KERR SPACETIME

In Boyer-Lindquist coordinates the Kerr metric is

$$\mathrm{d}s^2 = -\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\mathrm{d}t\,\mathrm{d}\phi + \frac{\Sigma}{\Delta}\mathrm{d}r^2 + \Sigma\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\mathrm{d}\phi^2, \tag{1}$$

where

$$\Sigma = r^2 + a^2\cos^2\theta, \quad \Delta = (r - r_+)(r - r_-), \tag{2}$$

$$A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta. \tag{3}$$

The horizon radii and angular velocity are

$$r_\pm = M \pm \sqrt{M^2 - a^2}, \qquad \Omega_H = \frac{a}{r_+^2 + a^2}. \tag{4}$$

We consider a weak cubic self-interaction

$$\Box\Phi + \epsilon|\Phi|^2\Phi = 0, \qquad \epsilon \in \mathbb{R}, \quad |\epsilon| \ll 1, \tag{5}$$

In four spacetime dimensions the canonical scalar has dimensions $[\Phi] = M^{-1}$, so ϵ is dimensionless in the action convention below. For a field amplitude that is not absorbed into the unit-horizon normalization, a useful local control parameter is

$$\chi_{\rm nl} = |\epsilon| M^2 \max_{\mathcal{D}} |\Phi^{(0)}|^2 \ll 1, \tag{6}$$

where $\mathcal{D}$ is the radial and angular region that dominates the cubic source projection. The reported $T^{(1)}$ and $R^{(1)}$ are coefficients in the derivative of the fixed-incident flux with respect to ϵ at $\epsilon = 0$; a finite-coupling application must additionally check that $|\epsilon T^{(1)}| \ll |T^{(0)}|$ and the analogous reflected-flux condition. We do not interpret the present frequency-domain calculation as a test of nonlinear saturation or of the expansion beyond this perturbative regime. This equation follows from the fixed-background scalar action

$$S_\Phi = \int \mathrm{d}^4x\,\sqrt{-g}\left[-g^{\mu\nu}\nabla_\mu\Phi^*\nabla_\nu\Phi + \frac{\epsilon}{2}|\Phi|^4\right]. \tag{7}$$

The stress tensor used below is the Hilbert stress tensor of the same effective field theory,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_\Phi}{\delta g^{\mu\nu}}. \tag{8}$$

For later use, the explicit tensor and the associated stationary Killing energy flux are

$$T_{\mu\nu} = \nabla_\mu\Phi^*\nabla_\nu\Phi \tag{9}$$

$$+ \nabla_\nu\Phi^*\nabla_\mu\Phi + g_{\mu\nu}\left[-\nabla_\alpha\Phi^*\nabla^\alpha\Phi + \frac{\epsilon}{2}|\Phi|^4\right], \tag{10}$$

$$j_E^\mu = -T^\mu{}_\nu\xi^\nu = -T^\mu{}_t, \qquad \xi = \partial_t,$$

$$\mathcal{F}_E^{\rm out}(r) = \int_0^{2\pi}\mathrm{d}\phi\int_0^\pi \mathrm{d}\theta\,\sqrt{-g}\,j_E^r, \tag{11}$$

$$\mathcal{F}_E^H = -\lim_{r\to r_+}\mathcal{F}_E^{\rm out}(r). \tag{12}$$

The reversed horizon orientation in the last definition makes positive $\mathcal{F}_E^H$ correspond to absorption when $p > 0$. The quartic term is proportional to $\delta^r{}_t$ in $T^r{}_t$ and therefore does not itself carry radial flux; the nonlinear boundary correction comes from the interference of the zeroth- and first-order kinetic terms. We now expand the field as

$$\Phi = \Phi^{(0)} + \epsilon\Phi^{(1)} + \mathcal{O}(\epsilon^2). \tag{13}$$

The zeroth-order incident wave is a single scalar spheroidal mode,

$$\Phi^{(0)} = e^{-\mathrm{i}\omega t + \mathrm{i}m\phi}S_{\ell m}(\theta; a\omega)R_{\ell m\omega}(r), \tag{14}$$

with angular normalization

$$\int |S_{\ell m}|^2\mathrm{d}\Omega = 1. \tag{15}$$

The scalar spheroidal harmonic obeys

$$\left[\frac{1}{\sin\theta}\partial_\theta\sin\theta\partial_\theta + a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta} + A_{\ell m}\right]S_{\ell m} = 0. \tag{16}$$

The radial Teukolsky equation for $s = 0$ is [32]

$$\mathcal{L}_{\ell m\omega} R \equiv \frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta \frac{\mathrm{d}R}{\mathrm{d}r}\right) + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right) R = 0, \quad (17)$$

where

$$K = (r^2 + a^2)\omega - am, \qquad \lambda_{\ell m\omega} = A_{\ell m} + a^2\omega^2 - 2am\omega. \quad (18)$$

The endpoint normalization used here can be related to the analytic MST solutions, which provide independent low-frequency connection formulae [30]. The numerical results below are assessed using internal spectral convergence and flux-balance diagnostics. The tortoise coordinate is defined by

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{r^2 + a^2}{\Delta}, \quad (19)$$

and the horizon wave number is

$$p = \omega - m\Omega_H. \quad (20)$$

The in solution is normalized to unit amplitude at the future horizon,

$$R^{\mathrm{in}}_{\ell m\omega} \sim e^{-\mathrm{i}pr_*}, \qquad r \to r_+, \quad (21)$$

and at infinity it has the form

$$R^{\mathrm{in}}_{\ell m\omega} \sim B^{\mathrm{inc}} \frac{e^{-\mathrm{i}\omega r_*}}{r} + B^{\mathrm{ref}} \frac{e^{+\mathrm{i}\omega r_*}}{r}, \qquad r \to \infty. \quad (22)$$

With this convention the linear flux coefficients are

$$T^{(0)} = \frac{p(r_+^2 + a^2)}{\omega |B^{\mathrm{inc}}|^2}, \qquad R^{(0)} = \frac{|B^{\mathrm{ref}}|^2}{|B^{\mathrm{inc}}|^2}. \quad (23)$$

For clarity, these coefficients are energy-flux ratios rather than merely amplitude ratios. Let $\xi = \partial_t$ be the stationary Killing field and let $T_{\mu\nu}$ be the scalar stress tensor defined in Eq. (8). The outward-oriented Killing energy current is $j^\mu_E = -T^\mu{}_t$, so at a surface of constant r we use $\mathcal{F}^{\mathrm{out}}_E = \int j^r_E \sqrt{-g}\, \mathrm{d}\theta \mathrm{d}\phi$. When quoting absorption by the future horizon, the horizon boundary orientation is reversed, so that positive absorption corresponds to positive $\mathcal{F}^H_E = -\mathcal{F}^{\mathrm{out}}_E|_{r=r_+}$ for $p > 0$. For a single Fourier mode, the energy current is the mode frequency times the radial Klein–Gordon current, up to one common positive normalization $\mathcal{N}$. Consequently,

$$\begin{aligned} \mathcal{F}^{\mathrm{in}}_{E,\infty} &= \mathcal{N}\omega^2 |B^{\mathrm{inc}}|^2, \\ \mathcal{F}^{\mathrm{ref}}_{E,\infty} &= \mathcal{N}\omega^2 |B^{\mathrm{ref}}|^2, \\ \mathcal{F}_{E,H} &= \mathcal{N}\omega p (r_+^2 + a^2). \end{aligned} \quad (24)$$

Their ratios give Eq. (23), including the sign of the horizon flux when $p < 0$. The self-interaction contributes a term proportional to $\delta^r{}_t$ in the mixed stress tensor and therefore does not add a radial boundary flux; at first order, the nonlinear flux correction is the interference term between $\Phi^{(0)}$ and $\Phi^{(1)}$ derived below.

For real ω the conserved radial current gives

$$T^{(0)} + R^{(0)} = 1. \quad (25)$$

At first order,

$$\Box \Phi^{(1)} = -|\Phi^{(0)}|^2 \Phi^{(0)}. \quad (26)$$

Since the source has the same (ω, m) dependence, we expand

$$\Phi^{(1)} = e^{-\mathrm{i}\omega t + \mathrm{i}m\phi} \sum_{\ell' \geq |m|} S_{\ell' m}(\theta; a\omega) R^{(1)}_{\ell'|\ell m\omega}(r). \quad (27)$$

After multiplying by Σ and projecting onto a target spheroidal harmonic, the radial source is

$$\mathcal{L}_{\ell' m\omega} R^{(1)}_{\ell'|\ell m\omega} = -Q_{\ell'\ell m}(r), \quad (28)$$

where

$$Q_{\ell'\ell m}(r) = \left[r^2 C^{(0)}_{\ell'\ell m} + a^2 C^{(2)}_{\ell'\ell m}\right] |R^{\mathrm{in}}_{\ell m\omega}|^2 R^{\mathrm{in}}_{\ell m\omega}, \quad (29)$$

and

$$C^{(j)}_{\ell'\ell m} = \int \mathrm{d}\Omega\, \cos^j\theta\, S^*_{\ell' m} |S_{\ell m}|^2 S_{\ell m}, \qquad j = 0, 2. \quad (30)$$

The self channel $\ell' = \ell$ is the only target channel that interferes with the linear outgoing wave in the total energy flux at order ϵ.

III. METHODS

A. Pseudo-spectral method for homogeneous solutions

The homogeneous radial equation is solved in the compact coordinate

$$z = \frac{r_+}{r}, \qquad z = 0 \text{ at infinity}, \qquad z = 1 \text{ at the horizon}. \quad (31)$$

As in the Schwarzschild calculation, the oscillatory endpoint phases are peeled off before Chebyshev collocation. This endpoint-first organization is also consistent with independent spectral black-hole perturbation work, where the asymptotic behavior is factored before the Chebyshev representation is truncated [31]:

$$R_{\mathrm{in}} = e^{-\mathrm{i}pr_*} u_{\mathrm{in}}(z), \quad (32)$$

$$R_{\mathrm{down}} = \frac{e^{-\mathrm{i}\omega r_*}}{r} u_{\mathrm{down}}(z), \quad (33)$$

$$R_{\mathrm{up}} = \frac{e^{+\mathrm{i}\omega r_*}}{r} u_{\mathrm{up}}(z). \quad (34)$$

The functions u_{in}, u_{down}, and u_{up} are smooth enough on their respective subdomains that the degenerate endpoint rows of the transformed ODE can be used directly as boundary regularity conditions. The outer and inner domains are matched at a finite radius $r_{\rm m}$ by continuity of R and $\mathrm{d}R/\mathrm{d}r$.

More explicitly, after the phase peeling the collocation equation has the form

$$B_2(z)u_{,zz} + B_1(z)u_{,z} + B_0(z)u = 0. \tag{35}$$

At the exact endpoint of each branch B_2 vanishes, so the endpoint row is the regular-singular relation $B_1 u_{,z} + B_0 u = 0$. The neighboring row is replaced by the normalization $u_{\rm down}(0) = u_{\rm up}(0) = 1$ on the outer domain or $u_{\rm in}(1) = 1$ on the inner domain. Thus the horizon is included as an exact collocation endpoint; no horizon truncation is introduced. Each domain uses $N+1$ Chebyshev points and a dense complex linear solve. The reference production runs use $N = 240$ on both domains, $r_{\rm m} = 40M$, Gauss order $q = 192$ for the compact inner integral, and relative tolerance 10^{-11} for the oscillatory phase-channel tail. The low-frequency runs use the same endpoint formulation with the sinh map described above; all reported results are obtained in IEEE double precision and are checked by the order, matching-radius, quadrature, Wronskian, and flux scans reported below.

The angular eigenvalue is computed by diagonalizing the truncated spherical harmonic representation of $\ell(\ell+1) - (a\omega)^2\cos^2\theta$, retaining sixteen modes above the requested ℓ. The source projectors are evaluated with normalized spheroidal harmonics and Gauss–Legendre quadrature. For the radial Green integrals, the compact inner part is integrated in z with the Jacobian $\mathrm{d}r = r_+/z^2\,\mathrm{d}z$, while the outer part is decomposed into its finite phase channels in r_* before oscillatory quadrature.

The scattering relation used in the code is the Kerr analogue of the Schwarzschild S-matrix relation,

$$R_{\rm in} = B^{\rm inc} R_{\rm down} + B^{\rm ref} R_{\rm up}. \tag{36}$$

Solving this two-by-two matching system gives $B^{\rm inc}$ and $B^{\rm ref}$ and therefore the linear coefficients in Eq. (23). The residual $|T^{(0)} + R^{(0)} - 1|$ is used as the first check of the homogeneous solve.

B. Nonlinear scattering coefficients

1. The nonlinear balance law

Let $R^{(\ell)}_{\rm up}$ denote the target-channel homogeneous solution that is purely outgoing at infinity. The Green-function solution in the self channel has outgoing amplitude at infinity and ingoing amplitude at the horizon. The conserved Wronskian is

$$W_\ell = \Delta\left(R^{\rm in}_{\ell m\omega}\frac{\mathrm{d}R^{\rm up}_{\ell m\omega}}{\mathrm{d}r} - R^{\rm up}_{\ell m\omega}\frac{\mathrm{d}R^{\rm in}_{\ell m\omega}}{\mathrm{d}r}\right), \tag{37}$$

which is independent of r for two homogeneous solutions of Eq. (17). The Green-function amplitudes are

$$A^{(1)}_{\rm ref} = -\frac{1}{W_\ell}\int_{r_+}^{\infty} Q_{\ell\ell m}(r) R^{\rm in}_{\ell m\omega}(r)\,\mathrm{d}r, \tag{38}$$

$$A^{(1)}_{H} = -\frac{1}{W_\ell}\int_{r_+}^{\infty} Q_{\ell\ell m}(r) R^{\rm up}_{\ell m\omega}(r)\,\mathrm{d}r, \tag{39}$$

where the minus sign follows from $\mathcal{L}_{\ell m\omega} R^{(1)} = -Q_{\ell\ell m}$. In the numerical implementation the Wronskian is evaluated from the matched homogeneous solutions and checked against its constancy. The amplitudes $A^{(1)}_{\rm ref}$ and $A^{(1)}_{H}$ are quoted in the unit-horizon normalization of Eq. (21). To form scattering coefficients one must instead hold the incident wave fixed. If the zeroth-order field is rescaled by $\alpha = 1/B^{\rm inc}$, the cubic source scales as $|\alpha|^2\alpha$. Therefore the fixed-incident first-order boundary amplitudes are

$$A^{(1)}_{\rm ref,fix} = \frac{A^{(1)}_{\rm ref}}{|B^{\rm inc}|^2 B^{\rm inc}}, \qquad A^{(1)}_{H,\rm fix} = \frac{A^{(1)}_{H}}{|B^{\rm inc}|^2 B^{\rm inc}}. \tag{40}$$

Expanding the horizon and infinity fluxes of the fixed-incident field then gives

$$T^{(1)} = \frac{2p(r_+^2 + a^2)}{\omega |B^{\rm inc}|^4}\,\mathrm{Re}\,A^{(1)}_{H}, \tag{41}$$

$$R^{(1)} = \frac{2\,\mathrm{Re}(\overline{B^{\rm ref}}\,A^{(1)}_{\rm ref})}{|B^{\rm inc}|^4}. \tag{42}$$

The current conservation identity gives the first-order balance law

$$T^{(1)} + R^{(1)} = 0. \tag{43}$$

Numerically this equation is the most useful global precision diagnostic, because it tests the homogeneous normalization, the nonlinear Green integral, and the flux normalization simultaneously. The fixed-incident normalization and the appearance of the $|B^{\rm inc}|^4$ denominators are derived explicitly in Appendix A.

2. Numerical accuracy diagnostics

The accuracy diagnostics used below are internal to the spectral calculation. Let the homogeneous equation be written at a collocation point as

$$\mathcal{R}_R = \Delta R'' + \Delta' R' + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right) R. \tag{44}$$

The reported relative residual is

$$\delta_R = \frac{|\mathcal{R}_R|}{|\Delta R''| + |\Delta' R'| + |(K^2/\Delta - \lambda_{\ell m\omega})\,R|}. \tag{45}$$

Thus the denominator contains the actual local field amplitude, not only the differential-equation coefficients.

This makes the residual insensitive to an overall rescaling of the homogeneous solution. The original radial residual is evaluated on the interior collocation nodes, $0 < z < 1$; the degenerate endpoint rows are reserved for the boundary regularity conditions and are not interpreted as ordinary pointwise ODE residuals. We also monitor the Chebyshev tail

$$\eta_{\rm tail} = \frac{\max_{N-15\le k\le N} |c_k|}{\max_{0\le k\le N} |c_k|}, \tag{46}$$

where c_k are the Chebyshev coefficients of the phase-peeled unknown on each subdomain. Finally, Eqs. (25) and (43) test the physical flux balance after the homogeneous matching and Green integral have both been performed.

The production data below retain the self channel $\ell' = \ell$ when quoting the first-order scattering coefficients. This is not an angular truncation of the first-order field: the cubic Kerr source in Eq. (29) does generate off-diagonal target channels. However, at order ϵ the total energy flux contains only interference between $\Phi^{(0)}$ and $\Phi^{(1)}$. Orthogonality of the spheroidal harmonics then removes every target channel except $\ell' = \ell$ from the flux correction. Off-diagonal channels are therefore required for the full first-order waveform, but not for the self-channel coefficients $T_\ell^{(1)}$ and $R_\ell^{(1)}$ reported here.

3. *Low-frequency behavior*

The Schwarzschild reduced radial equation admits a clean analytic low-frequency scaling argument. In the Kerr Teukolsky variable the far-zone radial normalization and the covariant source weight $r^2 C^{(0)}_{\ell\ell m} + a^2 C^{(2)}_{\ell\ell m}$ change the practical low-frequency behavior measured by the fixed-incident flux coefficients. Rather than importing the Schwarzschild scaling unchanged, we perform a direct Kerr low-frequency experiment for $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$.

The result is shown in Fig. 1. A log-log fit over the first four available low-frequency points gives

$$T_\ell^{(1)} \propto \omega^{\alpha_\ell}, \qquad \alpha_0 \simeq 2.50, \quad \alpha_1 \simeq 5.48, \quad \alpha_2 \simeq 8.74. \tag{47}$$

The corresponding linear slopes are approximately 2.34, 5.25, and 8.52. Thus, in the present Kerr normalization the nonlinear transmission correction follows the same centrifugal-barrier hierarchy as the linear transmission coefficient in the accessible low-frequency window. The reflected coefficient is harder to extract at the lowest frequencies because the infinity-side interference integral contains a large cancellation; this is why Eq. (43) is reported alongside the coefficients.

4. *High-frequency behavior*

At high frequencies the phase-peeled homogeneous amplitudes become slowly varying, while the nonlinear Green integral is controlled by rapidly oscillatory tails. The Schwarzschild calculation finds an exponential tail with a fitted temperature close to the Hawking temperature [33]. For Kerr we do not assume the Schwarzschild asymptotic integral unchanged, because the Teukolsky radial normalization and the Σ-weighted source modify the integrand. Instead we test the same one-parameter tail model,

$$T_\ell^{(1)}(\omega) \simeq A_\ell \exp\left[-\frac{\omega}{T_{\rm fit}}\right], \qquad \omega > {\rm Re}\,\omega_{\ell 0}. \tag{48}$$

For Kerr

$$T_H = \frac{r_+ - r_-}{4\pi(r_+^2 + a^2)}. \tag{49}$$

For $a = 0.5$ the fits in Sec. IV give $\pi M T_{\rm fit} = 0.112$, 0.104, and 0.100 for $\ell = 0, 1, 2$, close to $\pi M T_H = 0.1160$. These values should be read as finite-window effective temperatures rather than as a proof of an exact high-frequency theorem.

IV. RESULTS

Our main result on the Kerr nonlinear scattering coefficient is shown in Fig. 2. This figure is the direct Kerr analogue of the main Schwarzschild figure: it plots $T_\ell^{(1)}$ for $\ell = 0, 1, 2$, uses a linear scale to display the resonance peaks, and uses a logarithmic scale to display the high-frequency tail. The dashed lines mark the real parts of the fundamental scalar Kerr quasinormal modes computed with the local `qnm` package [34] for $s = 0$, $a = 0.5$, $m = 0$, and $n = 0$.

Three features are robust in the accepted data. First, in the axisymmetric Kerr cases studied here, $T_\ell^{(1)}$ is positive at every accepted sampled frequency in the present flux convention. For positive coupling in Eq. (5), the nonlinear correction therefore increases the absorbed flux and, by Eq. (43), decreases the reflected flux at the same order. This sign statement refers to the coefficient convention of Eqs. (41) and (42); it should not be compared to a reduced Schwarzschild field convention without translating the flux normalization.

Second, the peaks of $T_\ell^{(1)}$ track the frequency scale set by the real parts of the corresponding fundamental Kerr scalar QNMs [7, 34]. The fitted peak positions are $M\omega_{\rm peak} = 0.1299$, 0.3100, and 0.5022 for $\ell = 0, 1, 2$, while the QNM real parts are 0.1124, 0.2979, and 0.4920. The corresponding fractional offsets are approximately 16%, 4.1%, and 2.1%. This is the Kerr version of the Breit-Wigner-type resonance observed in the Schwarzschild nonlinear coefficients. The comparison is qualitative rather than a precision QNM determination:

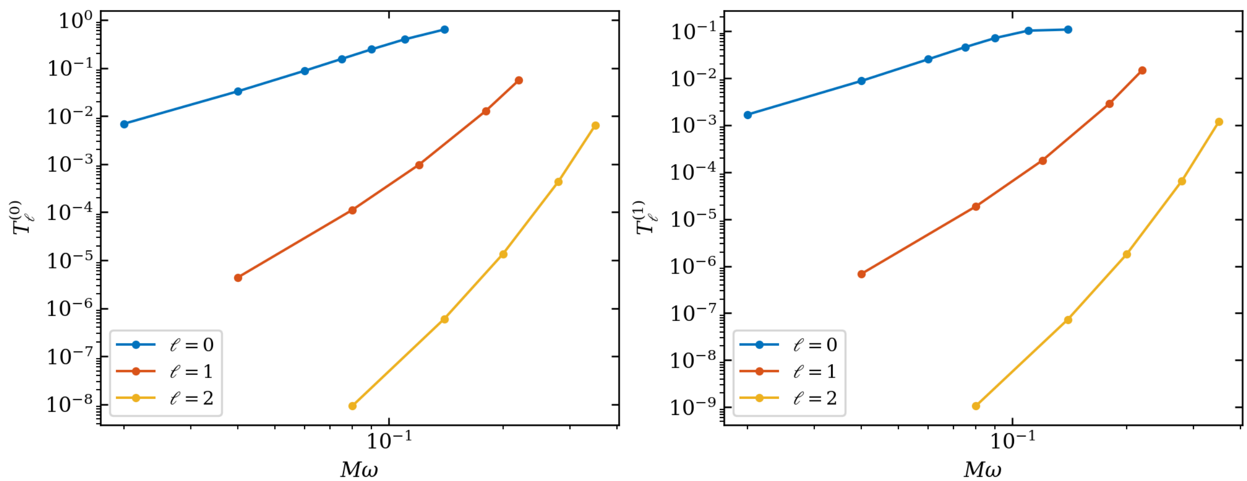

FIG. 1. Low-frequency Kerr diagnostics for $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$. Left: linear transmission coefficients. Right: self-channel nonlinear transmission coefficients. The fitted low-frequency slopes show that the Kerr fixed-incident coefficients decrease with the same centrifugal hierarchy in $T^{(0)}$ and $T^{(1)}$.

TABLE I. Fitting parameters of the Kerr nonlinear transmission resonance and the high-frequency Boltzmann tail. The QNM frequencies are scalar Kerr fundamental modes with $s = 0$, $a = 0.5$, $m = 0$, and $n = 0$. All frequencies are reported in units $M = 1$.

ℓ	ω_p	γ	$\Re\omega_Q$	$\|\Im\omega_Q\|$	$\pi T_{\rm fit}$	ΔT
0	0.1299	0.0779	0.1124	0.1022	0.1123	3.25%
1	0.3100	0.0597	0.2979	0.0954	0.1043	10.1%
2	0.5022	0.0611	0.4920	0.0946	0.1003	13.6%

the nonlinear coefficient is a real-frequency flux observable, whereas the QNM frequencies are poles of the analytically continued response. The quoted $\omega_{\rm peak}$ and γ are parameters of the numerical Lorentzian interpolation used to locate a broad feature; they are not a claim that the nonlinear flux has been analytically continued to a QNM pole. The fit uses the sampled points satisfying $T_\ell^{(1)} > 0.2T_{\ell,\rm max}^{(1)}$; the relative RMS residuals of these peak fits over the selected points are 1.36%, 1.23%, and 0.57% for $\ell = 0, 1, 2$, respectively. These residuals quantify the descriptive quality of the interpolation, not an error bar on the QNM frequencies.

Third, at high frequency the accepted tail points are well fit by the Boltzmann form Eq. (48). The fitted temperatures differ from the Kerr Hawking temperature by 3.2%, 10.1%, and 13.6% for $\ell = 0, 1, 2$, respectively. These are effective finite-window fit parameters, not an analytic derivation of a thermal spectrum. At low frequency the Kerr fixed-incident coefficients decrease with frequency in the accessible window. This differs from simply copying the Schwarzschild reduced-field statement and is therefore reported as a Kerr numerical result rather than assumed analytically.

The exponential fits use unweighted least squares for $\log T_\ell^{(1)}$ versus $M\omega$ over the windows $[0.18, 0.60]$, $[0.38, 0.85]$, and $[0.62, 1.00]$ for $\ell = 0, 1, 2$, respectively. The corresponding root-mean-square residuals in log space are 4.5×10^{-2}, 4.9×10^{-2}, and 3.8×10^{-2}. These window-dependent residuals quantify the quality of the finite-window fit; they are not estimates of a systematic error outside the computed frequency range.

For the accepted subset of the sweep, the largest first-order balance residual is 4.3×10^{-12}, 2.2×10^{-12}, and 1.2×10^{-12} for $\ell = 0, 1, 2$, respectively. The full sweep also contains rejected lowest-frequency points with residuals as large as 3.9×10^{-5} and 1.2×10^{-4} for $\ell = 1, 2$; these are caused by cancellation in the reflected interference term and are not used for precision claims. The transmitted coefficients remain smooth at those frequencies, so we retain them for the low-frequency slope measurement but explicitly separate that trend from an accuracy claim for $R^{(1)}$.

The change from the former production resolution to the active refined resolution is quantified separately in Table III. At the representative peak points the refined calculation changes the transmission coefficient by at most 3.2×10^{-5} and preserves the first-order flux balance at the 10^{-12} level. We therefore use the refined curves for the physical discussion below; the lower-resolution values are reported to make the resolution choice auditable rather than folded into an unreported systematic error.

We additionally test the self channel with $\ell = m = 2$ at both signs of the Kerr horizon frequency p and for a rapidly rotating case. Table IV records finite-resolution changes under N, $r_{\rm m}$, and q; these are not rigorous error bounds. The horizon amplitude is less sensitive than the reflected amplitude, while the Wronskian error remains below 6×10^{-16}. We therefore use flux balance, rather than one amplitude difference, as the acceptance criterion.

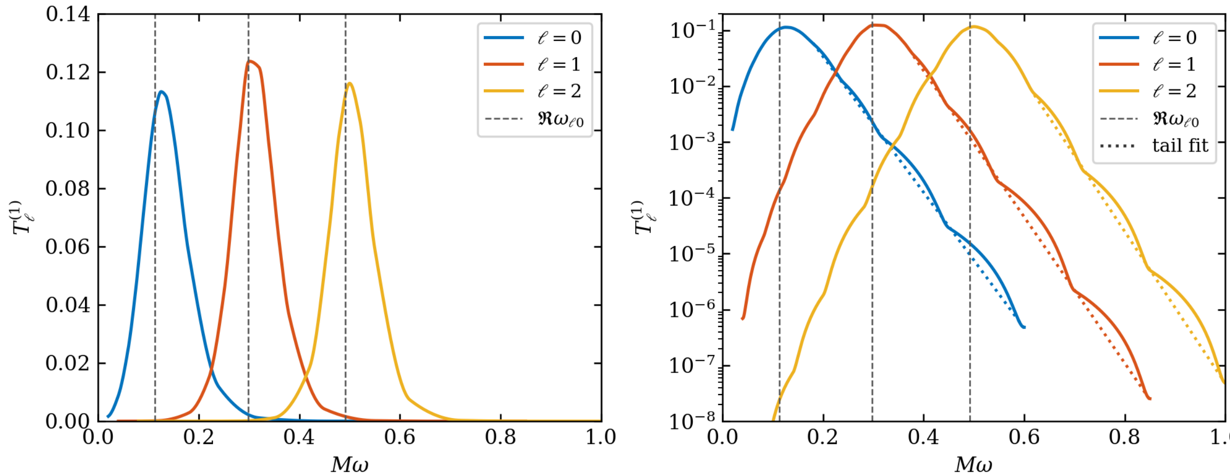

FIG. 2. Nonlinear Kerr transmission coefficients $T_\ell^{(1)}$ for $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$. The left panel uses a linear scale to show the Breit-Wigner-type peaks. The right panel uses a logarithmic scale to show the exponential high-frequency tails; dotted segments show the finite-window fits used in Table I. Dashed vertical lines mark the real parts of the fundamental scalar Kerr QNMs $\mathrm{Re}\,\omega_{\ell 0}$.

TABLE II. Numerical quality of the axisymmetric Kerr frequency sweeps. Here $\delta_0 = |T^{(0)} + R^{(0)} - 1|$ and $\delta_1 = |T^{(1)} + R^{(1)}|$. A point is counted as accepted when $\delta_1 \leq 10^{-8}$. All active frequency-sweep points use the refined resolution $N = 240$ and quadrature order $q = 192$; the earlier $N = 224$, $q = 160$ sweep is retained only as the production-resolution audit in Table III. The final column is the root-mean-square residual of the logarithmic high-frequency tail fit.

ℓ	points	$\max \delta_0$	accepted	$\max \delta_1^{\mathrm{acc}}$	σ_{tail}
0	16	2.2×10^{-11}	16/16	4.3×10^{-12}	4.5×10^{-2}
1	16	1.9×10^{-11}	14/16	2.2×10^{-12}	4.9×10^{-2}
2	15	3.8×10^{-11}	12/15	1.2×10^{-12}	3.8×10^{-2}

Local field diagnostic.
At the representative peak in Fig. 3, the field-dependent radial residual has maximum, 95th-percentile, and median values 2.5×10^{-10}, 3.3×10^{-12}, and 1.3×10^{-13}.
The largest normalized Chebyshev tail is 2.8×10^{-14} and the Wronskian mismatch is below 3×10^{-16}. These local diagnostics are reported separately from the global flux-balance residual.

A. Target-channel structure

The spheroidal source projection is a genuinely Kerr-specific part of the calculation. To expose it directly, we evaluated the target channels $\ell' = 2, \ldots, 6$ for the representative mode $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$ using $N = 288$, quadrature order $q = 224$, and Fourier-tail tolerance 10^{-11}. The amplitudes in Table V use the unit-horizon normalization of Eq. (21); they are not fixed-incident flux coefficients. Odd target channels vanish by the even-parity angular projectors in Eq. (29), while the even channels span several orders of magnitude in the outgoing Green amplitude.

The channel scan is converged for the dominant reflected amplitudes: changing N from 256 to 288 changes $A_{\mathrm{ref}}^{(1)}$ by 2.5×10^{-7} in the self channel and by 6.3×10^{-7} in the $\ell' = 4$ channel. The corresponding $\ell' = 4$ horizon amplitude changes by 7.1×10^{-3}, whereas the $\ell' = 6$ horizon amplitude is close to the double-precision cancellation floor and changes by order unity. For the displayed $\ell' = 6$ row, the independent diagnostic $(|B_{\mathrm{inc}}|^2 + |B_{\mathrm{ref}}|^2)/||B_{\mathrm{inc}}|^2 - |B_{\mathrm{ref}}|^2|$ is 3.27×10^{13}; this is the relevant conditioning indicator for the horizon-normalized continuation. We therefore use this experiment to establish the target-channel structure and the dominant outgoing amplitudes, not to claim uniform high precision for every off-diagonal horizon coefficient. Since off-diagonal channels do not interfere with the zeroth-order field, their energy flux first enters at $\mathcal{O}(\epsilon^2)$.

B. Non-axisymmetric superradiant check

The axisymmetric production sweep is useful for a controlled comparison with the Schwarzschild calculation, but it removes the most distinctive rotating black-hole effect because $p = \omega$ when $m = 0$. We therefore performed a bounded check with $a = 0.5$ and $\ell = m = 2$, using $N = 320$ on both spectral subdomains, quadra-

TABLE III. Production-resolution audit for the axisymmetric Kerr calculation. The entries compare the former production configuration $(N, q) = (224, 160)$ with the active reference configuration $(N, q) = (240, 192)$ at fixed $r_{\rm m} = 40$. Each entry in the first four columns is $|X_{240,192} - X_{224,160}|/|X_{240,192}|$ for the indicated Green-function amplitude or flux coefficient; the final column is the absolute first-order balance residual of the reference run.

case	$\Delta A_{\rm ref}$	ΔA_H	$\Delta T^{(1)}$	$\Delta R^{(1)}$	$\lvert\delta_1\rvert_{\rm ref}$
$\ell = 0,\ M\omega = .125$	4.5×10^{-7}	7.8×10^{-7}	2.3×10^{-6}	2.3×10^{-6}	2.7×10^{-12}
$\ell = 1,\ M\omega = .34$	5.9×10^{-6}	3.3×10^{-6}	1.5×10^{-5}	1.5×10^{-5}	2.2×10^{-12}
$\ell = 2,\ M\omega = .52$	9.6×10^{-6}	7.6×10^{-6}	3.1×10^{-5}	3.1×10^{-5}	1.2×10^{-12}

TABLE IV. Method-level convergence stress tests for the scalar Kerr Green integrals. Each entry is the maximum relative change in the indicated amplitude over the non-reference runs. The N scan compares $N = 224, 288$ with $N = 256$; the matching scan compares the listed alternatives with the reference $r_{\rm m}$; and the quadrature scan compares lower orders with $q = 192$. These non-axisymmetric self-channel tests are validation data for the numerical method and are not included in the axisymmetric production curves.

case	$\Delta A_{\rm ref}[N]$	$\Delta A_H[N]$	$\Delta A_{\rm ref}[r_{\rm m}]$	$\Delta A_H[r_{\rm m}]$	$\Delta A_{\rm ref}[q]$	$\Delta A_H[q]$	$\max \delta_W$
$a = .5,\ M\omega = .24$	2.1×10^{-5}	1.1×10^{-7}	3.3×10^{-5}	5.4×10^{-8}	2.8×10^{-8}	2.5×10^{-8}	5.9×10^{-16}
$a = .5,\ M\omega = .30$	3.8×10^{-5}	2.8×10^{-8}	2.0×10^{-5}	4.4×10^{-8}	1.8×10^{-6}	1.7×10^{-6}	3.7×10^{-16}
$a = .9,\ M\omega = .58$	1.5×10^{-6}	1.2×10^{-8}	1.7×10^{-6}	9.4×10^{-9}	1.4×10^{-10}	5.1×10^{-7}	2.9×10^{-16}

ture order $q = 320$, and $r_{\rm m} = 40M$. The threshold is $m\Omega_H = 0.2679491924$. The full ten-point scan is retained in the tracked CSV file; Table VI lists only points satisfying the same first-order balance gate $|T^{(1)} + R^{(1)}| \leq 10^{-8}$ used for the production data. The rejected points are $M\omega = 0.20, 0.22, 0.255, 0.265, 0.28$; they lie in the near-threshold portion of the scan and are limited by cancellation in the reflected interference term, rather than by a physical discontinuity.

Figure 4 shows the signed linear and first-order horizon flux coefficients. On the accepted points below the threshold, both $T^{(0)}$ and $T^{(1)}$ are negative, while they are positive on the accepted points above it. Thus the scalar Kerr calculation recovers the expected signed-flux change across the superradiant boundary in this representative mode. This is a validation of the horizon-frequency normalization, not a systematic survey over spin, m, or spheroidal target channels.

C. Kerr-to-Schwarzschild continuity

As a limiting-case check independent of external reference calculations, we hold $(\ell, m, M\omega) = (2, 0, 0.52)$ fixed and sweep the dimensionless spin from $a/M = 0$ to 0.5. The scalar spheroidal eigenvalue, the covariant source projection, the scattering amplitudes, and both flux coefficients are recomputed at every spin. At $a = 0$ the angular basis becomes spherical and the endpoint formulation reduces to the Schwarzschild spectral route; no radial endpoint is removed or replaced by a finite horizon cutoff. Figure 5 shows that $T^{(1)}$ and $-R^{(1)}$ vary smoothly from the $a = 0$ value, while Table VII records the balance and Wronskian diagnostics for the same points. The largest absolute first-order balance residual in this scan is 1.4×10^{-12} and the largest linear balance residual is below 1.2×10^{-11}. This experiment is a continuity and implementation check; it is not used as an external reference result or as a claim that the Kerr and Schwarzschild coefficient conventions are identical without the conversion in Appendix A.

V. DISCUSSION AND OUTLOOK

We have formulated the weakly nonlinear scattering problem for a scalar field on Kerr in a way that keeps the observable content of the Schwarzschild Green-function calculation unchanged. The linear in solution is normalized at the horizon, the incident normalization is restored through $B^{\rm inc}$, and the nonlinear corrections are read from the interference terms in the physical energy fluxes. In this form the two useful balance laws, $T^{(0)} + R^{(0)} = 1$ and $T^{(1)} + R^{(1)} = 0$, remain the main global checks of the calculation. The extension is not a formal replacement of the Schwarzschild radial variable by a Kerr one: the horizon frequency $p = \omega - m\Omega_H$, spheroidal angular basis, and Σ-weighted nonlinear source all enter the normalization of the observable coefficients.

For the axisymmetric Kerr sample $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$, the present numerical diagnostics support the following qualitative picture. The nonlinear transmission coefficient develops a single dominant Breit-Wigner-type peak for each ℓ, with the peak position tracking the frequency scale of the corresponding scalar Kerr quasinormal mode. The high-frequency tail is well fit by an exponential form in the computed window, and the fitted temperature is of the same order as the Kerr Hawking temperature. At low frequency, however, the fixed-incident Teukolsky normalization and the Kerr source weight pre-

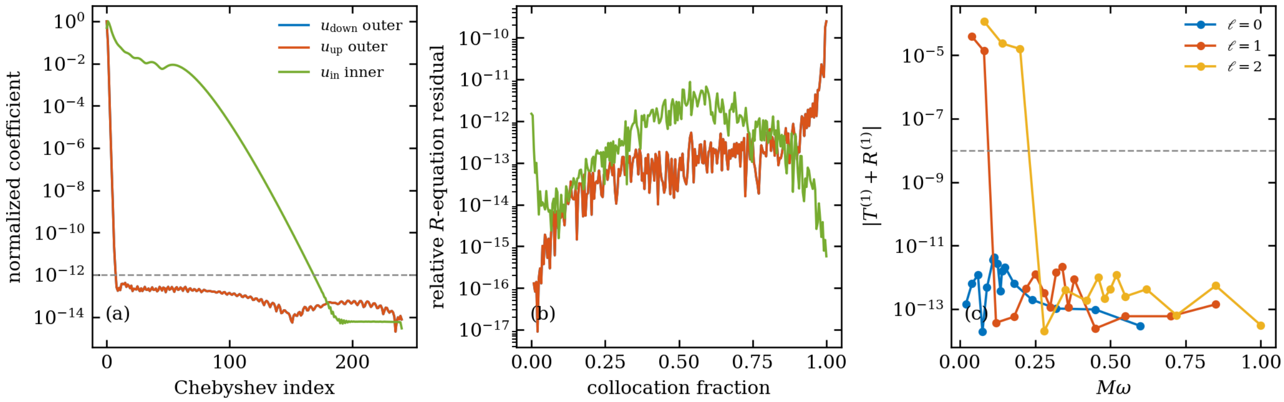

FIG. 3. Internal numerical accuracy diagnostics for the axisymmetric Kerr calculation. Panels (a) and (b) use the representative peak point $a = 0.5$, $m = 0$, $\ell = 2$, and $M\omega = 0.52$ with $N = 240$ and quadrature order $q = 192$. Panel (a) shows the normalized Chebyshev coefficients of the phase-peeled homogeneous solutions on the outer and inner subdomains. Panel (b) shows the relative residual (45) after substituting the numerical solution back into the original radial Teukolsky equation. Panel (c) shows the first-order flux-balance residual over the full axisymmetric frequency sweep. The dashed line marks the acceptance threshold 10^{-8} used in Table II.

TABLE V. Representative Kerr target-channel matrix for an incident $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$ mode. The angular entries are absolute values of the two source projectors in Eq. (29); the Green amplitudes are unit-horizon first-order amplitudes. The odd channels vanish within the displayed precision.

ℓ'	$\lvert C^{(0)}_{\ell'\ell m}\rvert$	$\lvert C^{(2)}_{\ell'\ell m}\rvert$	$\lvert A^{(1)}_{\mathrm{ref},\ell'\lvert\ell m}\rvert$	$\lvert A^{(1)}_{H,\ell'\lvert\ell m}\rvert$
2	1.136×10^{-1}	1.034×10^{-2}	1.137×10^{6}	3.804×10^{3}
3	0	0	0	0
4	3.578×10^{-2}	5.513×10^{-3}	1.362×10^{4}	1.319
5	0	0	0	0
6	5.341×10^{-3}	5.364×10^{-3}	2.265×10^{3}	5.383×10^{-7}

vent a direct transcription of the Schwarzschild reduced-variable scaling; the low-frequency behavior is therefore treated here as a numerical Kerr result.

The present data also identify the current limiting numerical effect. The transmitted nonlinear coefficient is smooth across the displayed low-frequency windows, while the reflected coefficient can suffer large cancellations in the interference integral. This is why the lowest-frequency $\ell = 1, 2$ points are used for the scaling trend but are not counted as high-precision reflected-flux points in Table II. The numerical evidence is therefore strongest for the resonance region, the high-frequency tail, and the transmitted low-frequency scaling; it is weaker for the individual low-frequency reflected coefficient before using the balance law. The representative target-channel experiment in Table V demonstrates the matrix structure generated by the Kerr source projection and shows that the dominant reflected off-diagonal amplitudes can be resolved at the stated spectral order. A broader scan over (a, ℓ, m, ω), uniform acceptance gates for off-diagonal horizon amplitudes, and a systematic study of the superradiant nonlinear flux remain future work.

The spin-limit audit supplies an additional implementation-level check. We fix $(\ell, m) = (2, 0)$ and $M\omega = 0.52$; the first-order coefficients change smoothly from the $a = 0$ endpoint to the production value at $a = 0.5$, while the two flux-balance laws remain satisfied. This continuity test does not replace a convention-matched Schwarzschild comparison, but it rules out a discontinuity introduced solely by the Kerr tortoise coordinate, spheroidal projection, or endpoint matching.

A. Scope and limitations

The conclusions above have a deliberately narrow interpretation. The calculation is a fixed-background scalar response problem with a prescribed cubic interaction; it is not a second-order metric perturbation, a gravitational self-force calculation, or an EMRI waveform model. The main frequency sweep is axisymmetric. The separate $m = 2$ check tests one narrow superradiant crossing but does not provide a systematic survey of superradiant sign changes. The target-channel experiment establishes the spheroidal coupling structure and the dominant reflected amplitudes, but the $\ell' = 6$ horizon coefficient is conditioning limited in double precision.

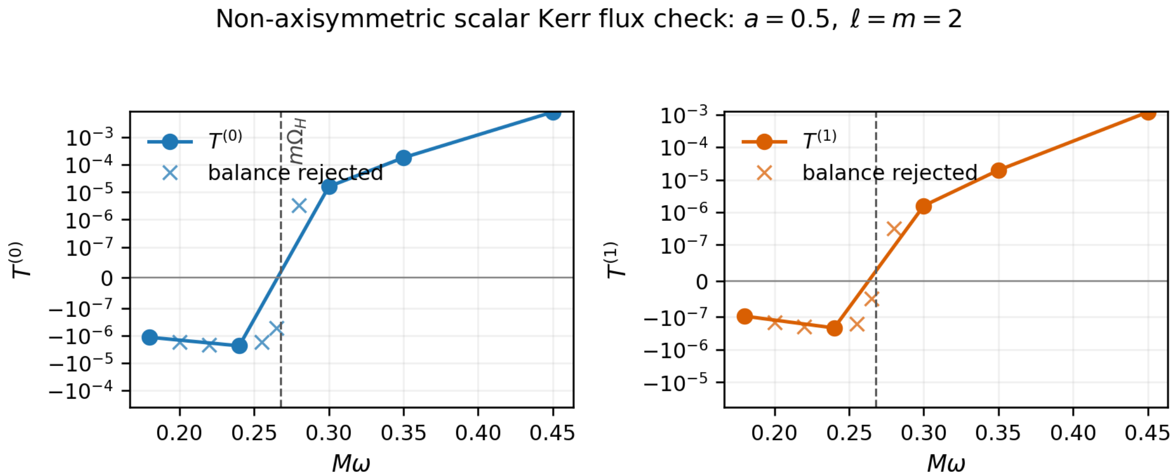

FIG. 4. Bounded non-axisymmetric scalar Kerr flux check for $a = 0.5$ and $\ell = m = 2$. The dashed line marks $M\omega = m\Omega_H = 0.2679491924$. Filled circles satisfy the first-order balance gate; crosses mark scan points retained in the CSV file but rejected because of cancellation. The signed coefficients change sign across the superradiant threshold on the accepted points.

TABLE VI. Accepted points in the bounded non-axisymmetric superradiant check. The entries are signed energy-flux coefficients for $a = 0.5$ and $\ell = m = 2$; $p = \omega - m\Omega_H$. The omitted points in the ten-point scan fail the 10^{-8} first-order balance gate.

$M\omega$	p	$T^{(0)}$	$T^{(1)}$	$\lvert T^{(1)} + R^{(1)}\rvert$
0.18	-8.795×10^{-2}	-1.157×10^{-6}	-9.742×10^{-8}	4.788×10^{-13}
0.24	-2.795×10^{-2}	-2.373×10^{-6}	-2.153×10^{-7}	1.261×10^{-12}
0.30	3.205×10^{-2}	1.595×10^{-5}	1.582×10^{-6}	9.240×10^{-13}
0.35	8.205×10^{-2}	1.804×10^{-4}	1.960×10^{-5}	1.598×10^{-12}
0.45	1.821×10^{-1}	8.298×10^{-3}	1.194×10^{-3}	4.130×10^{-12}

The QNM comparison uses real-frequency peak locations only, and the finite-window exponential fits should not be interpreted as QNM residues or as a proof of Hawking thermality. Extending the result to generic Kerr orbits or to gravitational perturbations requires, respectively, source-harmonic and orbit-evolution modules or metric reconstruction, gauge completion, and regularization; none of these are hidden in the present scalar result.

DATA AND CODE AVAILABILITY

The numerical data used in Figs. 1–5 and Tables I, II, III, IV, V, VI, and VII are included in the project data directory, "results/". The figures are generated by the scripts in "scripts/", and the exact commands and tracked inputs are listed in "docs/prd/README.md". The source code and frozen inputs are maintained at `https://github.com/ljq2088/KerrScattering`. The numerical checker compares the rounded entries in Tables I–V, the spin-limit audit, the production-resolution audit, and the low-frequency slopes with the tracked CSV files. The artifact manifest records file sizes and SHA-256 hashes for the figure and CSV inputs, including the target-channel values in Table V; a human-readable snapshot is provided in "docs/prd/reproducibility_report.md". No external experimental data are used.

Appendix A: Flux normalizations

1. Projected Kerr source

The scalar wave equation separates only after multiplying by $\Sigma = r^2 + a^2 \cos^2\theta$. At first order this gives

$$\Sigma\Box\Phi^{(1)} = -\Sigma|\Phi^{(0)}|^2\Phi^{(0)}. \tag{A1}$$

Using

$$\Phi^{(0)} = e^{-\mathrm{i}\omega t + \mathrm{i}m\phi} S_{\ell m} R_{\ell m\omega},$$
$$\Phi^{(1)} = e^{-\mathrm{i}\omega t + \mathrm{i}m\phi} \sum_{\ell'} S_{\ell' m} R^{(1)}_{\ell'|\ell m\omega}, \tag{A2}$$

and projecting with $S^*_{\ell' m}$ gives

$$\mathcal{L}_{\ell' m\omega} R^{(1)}_{\ell'|\ell m\omega} = -\left(r^2 C^{(0)}_{\ell'\ell m} + a^2 C^{(2)}_{\ell'\ell m}\right) |R^{\rm in}_{\ell m\omega}|^2 R^{\rm in}_{\ell m\omega}. \tag{A3}$$

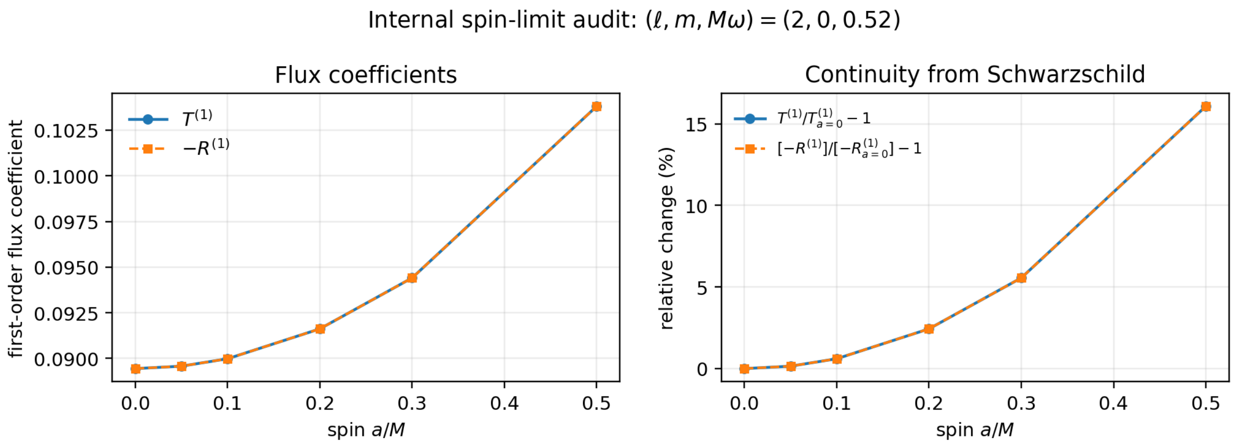

FIG. 5. Internal Kerr-to-Schwarzschild continuity test at fixed $(\ell, m, M\omega) = (2, 0, 0.52)$. The first-order transmission coefficient and $-R^{(1)}$ remain paired by the flux balance while changing smoothly with a/M. The right panel normalizes both coefficients to the $a = 0$ value; all points use the endpoint-inclusive spectral solver and are accepted by the first-order balance check.

TABLE VII. Internal Kerr-to-Schwarzschild spin-limit audit for $(\ell, m, M\omega) = (2, 0, 0.52)$. The entries use the same fixed-incident normalization as the production coefficients. The two balance columns are $T^{(0)} + R^{(0)} - 1$ and $T^{(1)} + R^{(1)}$, respectively.

a/M	$T^{(1)}$	$R^{(1)}$	$\lvert\delta_0\rvert$	$\lvert\delta_1\rvert$	$\lvert\delta_W\rvert$
0.00	8.94395×10^{-2}	-8.94395×10^{-2}	1.1×10^{-12}	3.8×10^{-13}	2.6×10^{-16}
0.05	8.95742×10^{-2}	-8.95742×10^{-2}	9.8×10^{-12}	1.1×10^{-12}	3.3×10^{-16}
0.10	8.99794×10^{-2}	-8.99794×10^{-2}	3.7×10^{-12}	4.9×10^{-13}	0
0.20	9.16178×10^{-2}	-9.16178×10^{-2}	4.3×10^{-13}	7.6×10^{-14}	1.8×10^{-16}
0.30	9.44095×10^{-2}	-9.44095×10^{-2}	1.1×10^{-11}	1.3×10^{-12}	1.8×10^{-16}
0.50	1.03814×10^{-1}	-1.03814×10^{-1}	6.0×10^{-12}	1.2×10^{-12}	0

Here

$$C^{(j)}_{\ell'\ell m} = \int \mathrm{d}\Omega\, \cos^j\theta\, S^*_{\ell' m} |S_{\ell m}|^2 S_{\ell m}, \qquad j = 0, 2. \quad \text{(A4)}$$

This is the point at which Kerr differs from the spherical Schwarzschild reduction: a single incident spheroidal channel sources a matrix of target channels with the same m. Only the target channel $\ell' = \ell$ contributes linearly to the total flux, because only that channel interferes with the zeroth-order outgoing field. Off-diagonal channels contribute to the energy flux first at order ϵ^2.

2. Linear current

The scalar radial current is

$$J[R] = \frac{\Delta}{2\mathrm{i}} \left(R^* \frac{\mathrm{d}R}{\mathrm{d}r} - R \frac{\mathrm{d}R^*}{\mathrm{d}r} \right). \quad \text{(A5)}$$

Using the asymptotics in Eq. (22), the far-zone current is

$$J_\infty = -\omega |B^{\mathrm{inc}}|^2 + \omega |B^{\mathrm{ref}}|^2. \quad \text{(A6)}$$

At the horizon, Eq. (21) gives

$$J_H = -p(r_+^2 + a^2). \quad \text{(A7)}$$

The equality $J_\infty = J_H$ gives Eq. (25).

3. Fixed-incident nonlinear flux

The Green-function amplitudes $A^{(1)}_{\mathrm{ref}}$ and $A^{(1)}_H$ are computed with the zeroth-order in solution normalized to unit horizon amplitude. Scattering coefficients, however, are defined at fixed incident amplitude. Therefore we rescale the zeroth-order solution by

$$\alpha = \frac{1}{B^{\mathrm{inc}}}. \quad \text{(A8)}$$

Because the source is cubic, the first-order field scales as $|\alpha|^2\alpha$. The fixed-incident self-channel field has the asymptotic form

$$R_{\mathrm{fix}} \sim \frac{e^{-\mathrm{i}\omega r_*}}{r} + \left(\frac{B^{\mathrm{ref}}}{B^{\mathrm{inc}}} + \epsilon \frac{A^{(1)}_{\mathrm{ref}}}{|B^{\mathrm{inc}}|^2 B^{\mathrm{inc}}} \right) \frac{e^{+\mathrm{i}\omega r_*}}{r}, \qquad r \to \infty, \quad \text{(A9)}$$

and

$$R_{\rm fix} \sim \left(\frac{1}{B^{\rm inc}} + \epsilon \frac{A_H^{(1)}}{|B^{\rm inc}|^2 B^{\rm inc}} \right) e^{-\mathrm{i}pr_*}, \qquad r \to r_+. \tag{A10}$$

The transmitted flux coefficient is therefore

$$\begin{aligned} T &= \frac{p(r_+^2+a^2)}{\omega} \left| \frac{1}{B^{\rm inc}} + \epsilon \frac{A_H^{(1)}}{|B^{\rm inc}|^2 B^{\rm inc}} \right|^2 + \mathcal{O}(\epsilon^2) \\ &= T^{(0)} + \epsilon \frac{2p(r_+^2+a^2)}{\omega |B^{\rm inc}|^4} \operatorname{Re} A_H^{(1)} + \mathcal{O}(\epsilon^2), \end{aligned} \tag{A11}$$

which gives Eq. (41). Similarly,

$$\begin{aligned} R &= \left| \frac{B^{\rm ref}}{B^{\rm inc}} + \epsilon \frac{A_{\rm ref}^{(1)}}{|B^{\rm inc}|^2 B^{\rm inc}} \right|^2 + \mathcal{O}(\epsilon^2) \\ &= R^{(0)} + \epsilon \frac{2 \operatorname{Re}(\overline{B^{\rm ref}}\, A_{\rm ref}^{(1)})}{|B^{\rm inc}|^4} + \mathcal{O}(\epsilon^2), \end{aligned} \tag{A12}$$

which gives Eq. (42). Conservation of the radial current for the inhomogeneous solution with no incoming first-order radiation then yields the first-order balance law Eq. (43).

4. Schwarzschild normalization correspondence

The numerical coefficients in this paper use unit incident amplitude. This point matters when comparing them with a reduced Schwarzschild calculation that quotes the first-order Green coefficient with the zeroth-order solution normalized at the horizon. In the nonrotating limit, write the scalar field in the reduced convention as $\Phi = e^{-\mathrm{i}\omega t} Y_{\ell m} u/r$ and let $\widehat{I}$ and $\widehat{R}$ denote the corresponding incident and reflected amplitudes. The present unit-horizon Teukolsky variable has $R^{\rm in} = r_+ u/r$ at zeroth order, so that

$$B^{\rm inc} = r_+ \widehat{I}, \qquad B^{\rm ref} = r_+ \widehat{R}. \tag{A13}$$

The cubic source is homogeneous of degree three. Consequently the horizon and infinity Green amplitudes obey $A_H^{(1)} = r_+^2 \widehat{A}_H^{(1)}$ and $A_{\rm ref}^{(1)} = r_+^3 \widehat{A}_{\rm ref}^{(1)}$ when the reduced solution is horizon normalized. For $a = 0$ (so $p = \omega$), Eqs. (41) and (42) therefore become

$$T_{\text{unit-inc}}^{(1)} = \frac{2 \operatorname{Re} \widehat{A}_H^{(1)}}{|\widehat{I}|^4}, \tag{A14}$$

$$R_{\text{unit-inc}}^{(1)} = \frac{2 \operatorname{Re}(\widehat{R}^* \widehat{A}_{\rm ref}^{(1)})}{|\widehat{I}|^4}. \tag{A15}$$

If the reduced Schwarzschild coefficients are instead defined with the common unit-horizon convention, $\widehat{T}^{(1)} = 2 \operatorname{Re} \widehat{A}_H^{(1)} / |\widehat{I}|^2$ and analogously for $\widehat{R}^{(1)}$, then

$$T_{\text{unit-inc}}^{(1)} = T^{(0)} \widehat{T}^{(1)}, \qquad R_{\text{unit-inc}}^{(1)} = T^{(0)} \widehat{R}^{(1)}. \tag{A16}$$

Thus a direct comparison of the numerical values of $T^{(1)}$ between the two papers is meaningful only after this normalization conversion. The physical first-order flux correction and the balance law are unchanged.

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